1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In bladder cancer, high stromal-to-tumor ratio (TSR) reflects TGF-β–driven fibroblast activation that creates a physical and immunosuppressive barrier, diminishing cytotoxic T cell infiltration and reducing immune checkpoint blockade (ICI) efficacy. MSI/dMMR-high tumors, by contrast, harbor elevated neoantigen loads that can partially overcome stromal exclusion, resulting in variable but potentially improved ICI responses. Therefore, a combined assessment of TSR (via TGFB1-driven signaling and stromal content) and MSI/dMMR status will identify subgroups with distinct ICI responses: stromal-high/MSI-stable tumors will be least responsive, stromal-high/MSI-high tumors will have intermediate responsiveness, and stromal-low tumors will show the strongest responses, with the potential for synergy if TGF-β pathway inhibitors are added to ICI in the stromal-high groups.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, MSI/dMMR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

MSI/dMMR: MSI/dMMR axis; KRT17 and MDK (EV/mRNA markers) and ADNP/PTGDS suggest tumor-intrinsic programs that correlate with aggressiveness and treatment resistance and could modulate immune context; SDC2/TFPI2 and urinary metabolites illustrate noninvasive readouts that could serve as orthogonal validators of stromal or tumor-state.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph

Secondary	MDK	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	ADNP	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective multi-omics analysis of bladder cancer cohorts to correlate TSR (from pathology and transcriptomic TGF-β signatures) and MSI/dMMR with ICI outcomes;

Prediction 2: spatial and single-cell profiling to map CAF subtypes, TGF-β activity, and immune cell localization;

Prediction 3: in vitro co-culture and organoid models manipulating TGF-β signaling and MSI status to test effects on T cell cytotoxicity;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- KRT17: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.